Project Design Phase-II: Technology Stack (Architecture & Stack)

Date:	30 September 2026
Team ID:	[SWTID-2026-3230]
Project Name:	Auto Ticket Classification using Flow Designer
Maximum Marks:	4 Marks

Technical Architecture

The deliverable includes the technical architecture flow for Auto Ticket Classification using ServiceNow Flow Designer as described below:

1. User Submission: End-user (Student/Faculty) creates an incident ticket via the ServiceNow Service Portal by entering a short description and detailed description.
2. Event Trigger: Database insert trigger invokes the ServiceNow Flow Designer workflow upon new incident record creation.
3. NLP & Pattern Triage: Flow Designer executes custom scripts / Integration Hub to parse text, extract keywords, and match patterns.
4. Automated Assignment: The ticket is automatically categorized, prioritized, and routed to the correct Assignment Group (e.g., Network, Hardware, Software).
5. Notification & Dashboard: System updates the Incident record, sends email alerts, and populates the support agent executive dashboard.

Table-1 : Components & Technologies

S.No	Component	Description	Technology
1	User Interface	How user interacts with application via Service Portal, Web UI, and Dashboards	HTML, CSS, JavaScript, ServiceNow Service Portal UI
2	Application Logic-1	Incident record handling, form scripts, and client/server business logic	JavaScript, ServiceNow Server Scripts (Script Includes)

3	Application Logic-2	Automated workflow orchestration, conditional branching, and decision tables	ServiceNow Flow Designer & Decision Tables
4	Application Logic-3	Text analysis, keyword extraction, and pattern matching for categorization	Regular Expressions / ServiceNow Predictive Intelligence
5	Database	Relational database tables storing incident records, user profiles, and logs	ServiceNow CMDB & Task Database (MariaDB / MySQL backend)
6	Cloud Database	Enterprise cloud data storage and transactional data persistence	ServiceNow Instance Cloud Storage
7	File Storage	Attachment storage for screenshots, error logs, and supporting documents	ServiceNow Attachments Document Store
8	External API-1	Notification services for email and push alerts	ServiceNow Notification Email Engine / SMTP
9	External API-2	Integration endpoints for external authentication or SSO	OAuth 2.0 / SAML 2.0 Single Sign-On API
10	Machine Learning Model	Classification engine for predicting category and subcategory	ServiceNow Predictive Intelligence Classification Solution
11	Infrastructure (Server / Cloud)	Cloud deployment infrastructure managed on ServiceNow enterprise platform Local Server: Developer Instance Cloud Server:	ServiceNow PaaS Cloud Infrastructure

			ServiceNow Multi-Tenant Cloud Architecture

Table-2: Application Characteristics

S.No	Characteristics	Description	Technology
1	Open-Source Frameworks	Open web technologies and libraries utilized in portal UI development	Bootstrap, AngularJS / Angular, jQuery
2	Security Implementations	Access controls, encryption standards, and security hardening	Role-Based Access Control (RBAC), ACLs, HTTPS/TLS 1.3, OAuth 2.0, OWASP guidelines
3	Scalable Architecture	Multi-tier enterprise cloud architecture with decoupled workflows and asynchronous job execution	ServiceNow Flow Execution Engine, Asynchronous Business Rules, Multi-Node Cloud Architecture
4	Availability	High availability design utilizing cloud redundancy, failover mechanisms, and load balancers	ServiceNow High Availability (HA) Cloud Infrastructure, Global Load Balancers
5	Performance	Optimization for low latency, rapid ticket triage, and fast response times under high concurrency	Instance Caching, Asynchronous Flow Worker Queues, Optimized Database Indexing